appear to slow down as it approached the horizon, becoming redder and dimmer along the way. They wouldn't ever see it cross; to someone far away, objects appear to be frozen in time as they approach the horizon, rather than plunging in. This led astrophysicists to develop a picture called the *membrane paradigm*, according to which we can model the physical properties of a black hole by imagining that there is a physical membrane at the horizon, with certain calculable properties such as temperature and electrical conductivity. The membrane paradigm was originally thought of as a convenient shortcut through which astrophysicists could simplify calculations involving black holes, but complementarity claims that external observers really do see black holes as if they were vibrating quantum membranes where the classical event horizon would be.

If you tend to think of spacetime as a fundamental thing, this might make no sense at all. Spacetime has some geometry, there's nothing else to it. But quantum-mechanically it's perfectly plausible; there's a wave function of the universe, and different observations can reveal different things about it. It's not that much different from saying that the number of particles in a state depends on how we observe it.

The world is a quantum state evolving in Hilbert space, and physical space emerges out of that. It shouldn't come as a surprise that a single quantum state might exhibit different notions of position and locality depending on what kind of observations we perform on it. According to black-hole complementarity, there's no such thing as "what the geometry of spacetime is," or "where the degrees of freedom are"; you ask either what the quantum state is, or what is seen by some particular observer.

This sounds different from the picture we explored in the last chapter, where degrees of freedom were distributed in a network filling space, and became entangled to define an emergent geometry. But that picture was only meant to apply when gravity was weak, and black holes definitely do not qualify as weak. In the view presented in this chapter, there are still abstract degrees of freedom coming together to form spacetime, but "where they are located" depends on how they are being observed. Space itself is not fundamental; it's just a useful way of talking from certain points of view.